

C05: Assessment Tool-1

1. Design pseudocode to analyze a social network graph

(a) Find Connected Components:

START

Input number of users N

Input number of connections between users

Create a graph G

Mark all users as NOT VISITED

For each user U from 1 to N ~~if~~

IF U IS NOT VISITED THEN

Perform DFS(U)

END IF

END FOR

DFS(U)

Mark U as VISITED

Print U

FOR each user V connected
to U

IF V IS NOT VISITED

THEN

DFS(V)

END IF

END FOR

STOP

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(b) Suggest shortest path:

START

Input graph G

Input source user S

Input destination user D

Mark all users as NOT VISITED

Create an empty queue

Add S to queue

Mark S as VISITED

WHILE queue is not empty

Remove user U from queue

IF $U = D$ THEN

Print "Shortest path found"

STOP

END IF

FOR each user V connected to U

IF V IS NOT VISITED THEN

Mark V as VISITED

Add V to queue

END IF

END FOR

END WHILE

Print "No path exists"

STOP

Project Task dependencies:

START

Input no. of tasks N

Input task dependencies

create a directed graph G

Find the in-degree of each task

Create an empty queue

FOR each task T

IF in-degree $[T] = 0$ THEN

Add T to queue

END IF

END FOR

While queue is not empty

Remove task T from queue

Point T

FOR each task V connected from T

Decrease in-degree $[V]$ by 1

IF in-degree $[V] = 0$

THEN

Add V to queue

END IF

END FOR

END WHILE

STOP